# SOP-MEASURING & MONITORING OF EQUIPMENTS

[CONFIDENTIAL]

[UTTHUNGA TECHNOLOGIES]
[IndiQube, South Summit, South End Rd, Vijayarangam Layout, Bengaluru, Karnataka 560004]

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## CHANGE HISTORY

This section describes the details of changes that have resulted in the current document.

| SL. No. | Document Version No. | Change Description | Reason For change | Prepared By & Date | Reviewed By & Date | Approved By & Date |
|---|---|---|---|---|---|---|
| 01 | Issue B Version 0.1 | Implementation of the Integrated Management System | ISO 9001:2015 & ISO 80079-34 Ed.2 requirement are incorporated | Vinothkumar. P.S 12-Dec-2025 | Santosh V 12-Dec-2025 | |
| 02 | Issue B Version 1.0 | | | | | Sanmangala M S 12-Dec-2025 |

# 1 Purpose

To ensure that all measuring and monitoring equipment (MME) used for product conformity, including Ex-equipment covered under ISO/IEC 80079-34—is calibrated, controlled, and maintained to ensure valid and reliable measurement results, in compliance with:

a. ISO 9001:2015 Clause 7.1.5
b. ISO/IEC 80079-34:2018 requirements for control of inspection, measuring and test equipment (IMTE), traceability, and documentation.

# 2 Scope

This procedure applies to all MME used for:

a. Inspection and testing of materials, components, sub-assemblies, and final products.
b. Ex-specific measurements (e.g., temperature, pressure, clearances, electrical tests) relevant to conformity with type-examination certificates (ATEX/IECEx).
c. Any measurement influencing product quality or safety.

# 3 Abbreviations

a. **MME:** Measuring and Monitoring Equipment
b. **IMTE:** Inspection, Measuring & Test Equipment.
c. **OOT:** Out of Tolerance

# 4 Terms and Definitions

a. **Calibration:** Comparison of MME against a standard with known accuracy.
b. **Adjustment:** Corrective action to bring MME into specification.
c. **Traceability:** Unbroken chain of calibrations referenced to national/international standards.

# 5 Responsibilities

| Process | Responsible |
|---|---|
| Quality Assurance Manager | Ensure compliance with ISO 9001 & ISO/IEC 80079-34 calibration requirements; approve calibration suppliers. |
| Quality Control | Maintain MME register; schedule calibrations; quarantine out-of-tolerance (OOT) equipment. |

| | |
|---|---|
| Production Supervisors | Ensure required MME is calibrated and available before use. |
| Users/Technicians /Operator | Inspect equipment before use; report damage or suspected inaccuracies. |

# 6 RACI

| Process | Responsible | Accountable | Consulted | Informed |
|---|---|---|---|---|
| Control of Measuring and monitoring of equipment's | Quality Control | Quality Assurance Manager | Quality Director | Production/ Customer/ Top Management |

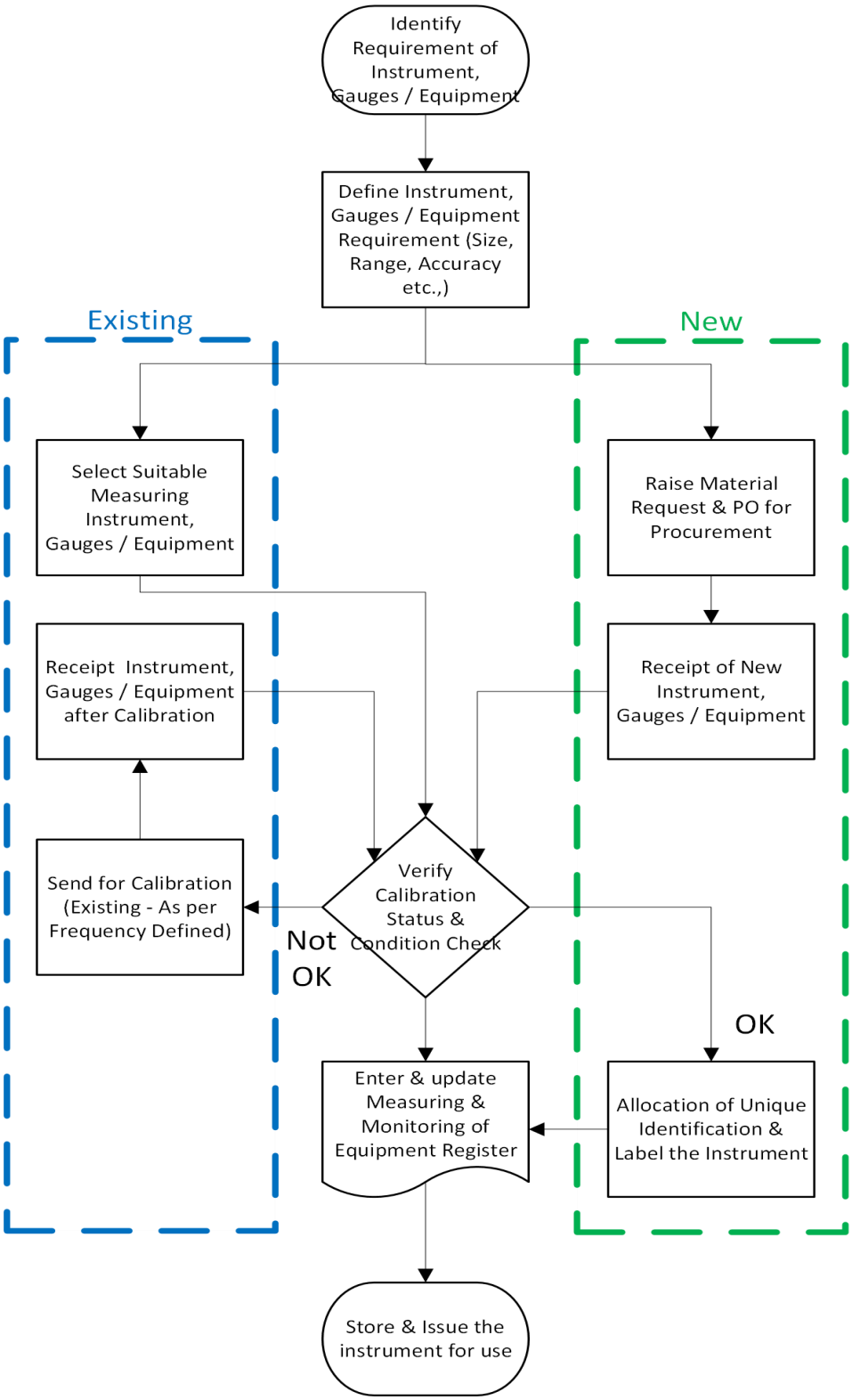
# 7 SIPOC

## Control of Measuring and Monitoring Equipment

| Supplier | Input | Process and control | | Output | Customer |
|---|---|---|---|---|---|
| | | Process | Control | | |
| Supplier / Calibration agency / | Standard / MME Register / | Control of measuring & monitoring equipment | Review of MME Register | Calibration report / MME Register / Masters Calibration report / OOT | Calibration Report |

# 8 Overall Process Flow Chart

## a. Monitoring & Measuring Equipment (Calibration Process)



# 9 Process Description

## 9.1 Identify Requirement

Identify the required measurement or monitoring instrument, gauges or equipment are available as per the product / process quality requirement

If not available, it needs to be ordered, by consideration of the following criteria,

a. Verify suitable measuring instruments, gauges / equipment's are available to measure the product character,
b. Control manufacturing processes
c. Ensure Ex-critical characteristics are met
d. Demonstrate compliance with standards, drawings, and specifications

## 9.2 Define & Selection of MME

Defining the measuring or monitoring instrument, gauges or equipment for each characteristic, need to consider the following criteria as follows, before procurement & for selection of measurement.

a. **Measurement range** – must cover the expected values
b. **Accuracy & uncertainty** – the accuracy (or uncertainty) of the measuring instrument should be at least 10 times better than the tolerance of the parameter being measured.
c. **Resolution / sensitivity** – appropriate for the parameter measured
d. **Type of equipment** – gauge, meter, tester, sensor, software, etc.
e. **Environmental suitability** – temperature, humidity, vibration, Ex area if applicable
f. **Traceability requirements** – calibration traceable to national/international standards
g. **Frequency of use & calibration interval**
h. **Ex relevance** – whether equipment is used to verify Ex-critical characteristics

## 9.3 Verify Calibration Requirements (Status & Condition)

Calibrations verification shall be performed as follows:

a. An accredited NABL laboratory (preferably ISO/IEC 17025),
b. Calibration shall be traceable to national/international standards (e.g., NIST, PTB).
c. If no traceable standard exists, define and document an internal method (ISO 9001:2015 compliance requirement).
d. Calibration intervals shall be defined based on any of the following:

e. Manufacturer recommendations
f. Historical stability
g. Frequency of use
h. Criticality to Ex-safety (ISO/IEC 80079-34 requirement)
i. Ensure no damage

## 9.4 Calibration Record (MME Register)

Maintain a Master MME Register with:

a. Unique equipment ID
b. Serial number
c. MME Name (Equipment Name)
d. Model (Equipment Model)
e. Equipment installed in location,
f. Equipment used in process,
g. Calibration frequency
h. Equipment calibration location Inhouse / Outsourced,
i. Equipment calibration company name (Calibration agency)
j. Calibration method/specification
k. Equipment range
l. Equipment operating range
m. Margin of tolerance (Accuracy)
n. Calibration point
o. Calibration report number and date,
p. Calibration result,
q. Calibration report & label verified by name
r. Next due date
s. Data entered by
t. Data verified by

## 9.5 Labelling

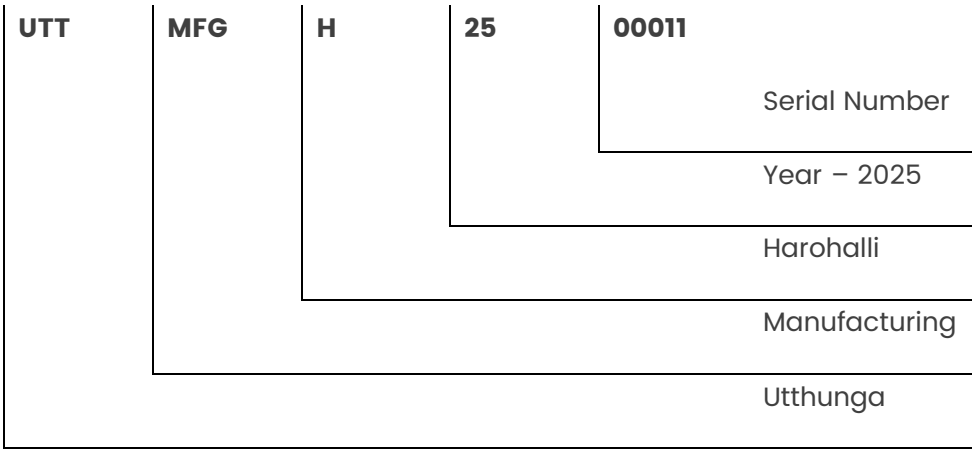
Attach a calibration label including:

a. Equipment ID
b. Calibration date
c. Next calibration due date
d. Calibration organization

For Ex-equipment control (per ISO/IEC 80079-34), labelling must be legible, tamper-resistant, and placed so as not to obscure safety markings.

### 9.6 Unique Identification Call Out Standard

Example – Unique ID #: **UTTMFGH2500011**




### 9.7 Out-of-Tolerance (OOT) Conditions

If equipment fails calibration:

a. Quarantine the device immediately.
b. Identify all processes or products potentially affected.
c. Perform impact analysis to determine:
    - Validity of previous measurements
    - Potential need for re-inspection or rework
d. Record OOT investigation results.
e. Notify Quality Assurance Manager for final disposition.
f. Document actions per Nonconformance Procedure.

### 9.8 Use & Handling of MME

1. Users shall perform pre-use checks (damage, cleanliness, battery, zero-setting).
2. Protect MME from environmental conditions that may affect accuracy.
3. Store delicate instruments in protective cases.
4. Do not use uncalibrated or expired MME for product acceptance.
5. Ensure labelling are available and legible

### 9.9 Maintenance & Repair

1. MME requiring repair must be:
   a. Rejection (Red) Tag denotes "OUT OF SERVICE"
   b. Removed from production use
2. After repair, calibration is required before use.

### 9.10 Documentation Requirements

Document control for measuring and monitoring equipment are maintained as follows:

a. Calibration certificates from NABL laboratory
b. MME register
c. OOT reports
d. Traceability documentation (Master Equipment)

Records shall be retained for a minimum of 10 years or per customer/regulatory requirements.

## 10 Information Security requirements

a. Information Security incidents and complaints are logged and closed appropriately
b. Confidentiality of information and project details is maintained by the team
c. Archiving and back up of data as per contractual obligations

## 11 Metrics

a. On time calibration

## 12 References

a. ISO 9001 Clauses 4.2, 8.2, 8.5, 9.1
b. ISO 80079-34:2018 Clauses 4.2, 8.2, 8.5, 9.1
c. ISO 27001 Clauses 4.2, 8.1, Controls A.16
d. ISMS Control Policies
e. ISO/IEC 17025

## 13 Annexures

a. NIL

## 14 Forms

a. MME Register – UTTMFGMMTMP19